

Vershynin's exponential singularity term is resolved in the i.i.d. subgaussian symmetric model

Literature-resolution packet for arXiv:1102.0300

Status: verified later-literature answer, not a new result. The canonical i.i.d. formulation is Theorem 1.1 of arXiv:2203.06141v3.

1 The question isolated

Let H be an $n \times n$ real symmetric random matrix with independent subgaussian entries on and above the diagonal, of mean zero and variance one. The 2011 paper of Vershynin proved, in a somewhat more general model, an estimate of the form

$$\mathbb{P}\left\{\sigma_{\min}(H) \leq \varepsilon n^{-1/2}\right\} \leq C\varepsilon^{1/9} + 2e^{-n^c}.$$

Its Remark 1.4 conjectured that the singularity-probability term $2e^{-n^c}$ could be replaced by $2e^{-cn}$. The surrounding discussion also identifies linear dependence on ε as the optimal small-ball scale.

While Nguyen's results give weaker conclusions than the results in this paper, they hold under somewhat weaker conditions on the distribution than **(H)** (for example, the entries of H do not to have mean zero); see [11] for precise statements.

Remark 1.4 (Optimality). Although the magnitude of the gap $n^{-1/2}$ in Theorem 1.1 is optimal, the form of (1.1) and (1.8) suggests that the exponent $1/9$ is not optimal. Indeed, our argument automatically yields $\varepsilon^{1/8+\delta}$ for every $\delta > 0$ (with constants C, c depending also on δ). Some further improvement of the exponent may be possible with a more accurate argument, but the technique of this paper would still not reach the optimal exponent 1 (in particular, due to losses in decoupling). Furthermore, we conjecture that the singularity probability term $2e^{-n^c}$ in (1.5) may be improved to $2e^{-cn}$.

1.3 Four moments

Even without subgaussian assumption (1.4) on the entries of H , the bound on the spectral norm $\|H\| = \max_k |\lambda_k(H)|$ can be removed from (1.3), however this will lead to a weaker probability bound than in Theorem 1.2

2 The later theorem

Campos, Jenssen, Michelen and Sahasrabudhe prove the following in arXiv:2203.06141v3, Theorem 1.1.

Resolving theorem. Let ζ be a subgaussian random variable with mean zero and variance one, and let A be a symmetric $n \times n$ matrix whose entries on and above the diagonal are independent copies of ζ . There are constants $C, c > 0$, depending only on the law of ζ , such that for every $\varepsilon \geq 0$,

$$\mathbb{P}\left\{\sigma_{\min}(A) \leq \varepsilon n^{-1/2}\right\} \leq C\varepsilon + e^{-cn}. \quad (1)$$

The resolving paper explicitly says that this confirms the folklore conjecture stated by Vershynin and is sharp up to the constants. Thus the identification is not merely an inference based on similar-looking estimates.

Setting $\varepsilon = 0$ in (1) gives

$$\mathbb{P}\{A \text{ is singular}\} \leq e^{-cn},$$

which is precisely the conjectured exponential-in- n improvement in the canonical i.i.d. model. Equation (1) is stronger: it also upgrades $\varepsilon^{1/9}$ to the optimal linear dependence ε . The Rademacher law is subgaussian, so symmetric Bernoulli matrices are included.

Wigner matrix ensemble, was introduced in the 1950s in the seminal work of Wigner [20], who established the famous “semi-circular law” for the eigenvalues of such matrices.

In this paper we study the extreme behavior of the *least singular value* of A , which we denote by $\sigma_{\min}(A)$. Heuristically, we expect that $\sigma_{\min}(A) = \Theta(n^{-1/2})$ and thus it is natural to consider

$$\mathbb{P}(\sigma_{\min}(A) \leq \varepsilon n^{-1/2}), \quad (1)$$

for all $\varepsilon \geq 0$ (see Section 1.2). In this paper we prove a bound on this quantity which is optimal up to constants, for all random symmetric matrices with i.i.d. *subgaussian* entries. This confirms the folklore conjecture, explicitly stated by Vershynin in [46].

Theorem 1.1. *Let ζ be a subgaussian random variable with mean 0 and variance 1 and let A be a $n \times n$ random symmetric matrix whose entries above the diagonal $(A_{i,j})_{i \leq j}$ are independent and distributed according to ζ . Then for every $\varepsilon \geq 0$,*

$$\mathbb{P}_A(\sigma_{\min}(A) \leq \varepsilon n^{-1/2}) \leq C\varepsilon + e^{-cn}, \quad (2)$$

where $C, c > 0$ depend only on ζ .

This conjecture is sharp up to the value of the constants $C, c > 0$ and resolves the “up-to-constants” analogue of the Spielman–Teng conjecture for random symmetric matrices (see Section 1.2). Also note that the special case $\varepsilon = 0$ tells us that the singularity probability of any random symmetric A with subgaussian entry distribution is exponentially small, generalizing our previous work [4] on the $\{-1, 1\}$ case.

3 Scope and logical crosswalk

The implication is immediate and contains no hidden limit argument:

$$\text{Theorem 1.1 of arXiv:2203.06141} \xRightarrow{\varepsilon=0} \text{Vershynin's } e^{-cn} \text{ singularity term.}$$

There are two scope distinctions worth preserving.

1. Vershynin’s hypothesis **(H)** allowed the diagonal entries to be arbitrary (subject in Theorem 1.2 to a size bound), independently of the off-diagonal entries. The resolving theorem assumes the entries on and above the diagonal are i.i.d. This packet therefore certifies the full canonical i.i.d. subgaussian formulation—the one the resolving authors themselves call Vershynin’s conjecture—and does not assert an arbitrary-diagonal extension.
2. The older introduction also recalls the sharper conjecture that a symmetric Bernoulli matrix has singularity probability $(1/2 + o(1))^n$. An unspecified e^{-cn} upper bound does not determine that optimal exponential base. That separate conjecture is not claimed solved here.

These qualifications do not weaken the answer to the queue’s exact “we conjecture” signal in Remark 1.4 as interpreted in the later paper; they prevent conflating it with stronger or more uniform variants.

4 Provenance and verification

The source is R. Vershynin, *Invertibility of symmetric random matrices*, arXiv:1102.0300v4, Random Structures & Algorithms 44 (2014), 135–182. The resolving paper is M. Campos, M. Jenssen, M. Michelen and J. Sahasrabudhe, *The least singular value of a random symmetric matrix*, arXiv:2203.06141v3, accepted in Forum of Mathematics, Pi.

The packet includes the official current arXiv PDFs, exact source and theorem crops, and a separate verification record. No part of the mathematical proof is attributed to this run.